Lecture 2. Data Visualization

Why visualise, taxonomy of data graphics, a directory of plots, distributions, associations, trends, telling a story

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These notes follow the course slides (based on *Modern Data Science with R*, Baumer, Kaplan and Horton, and Wilke's *Fundamentals of Data Visualization*). Lines starting with ## are R output; data sets come from R packages. Material marked [Supplement] is not on the slides.

2.1 Why visualise data

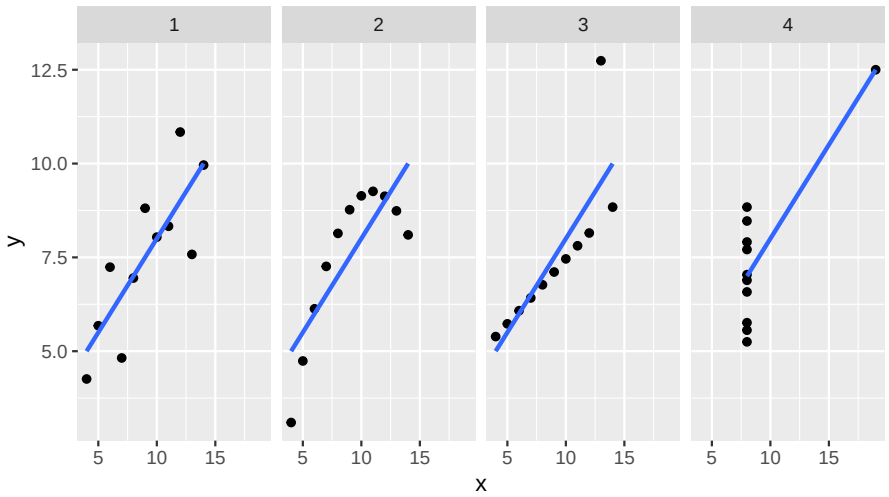
Motivation. Very different data sets can share the same summary statistics; a plot shows what the numbers hide.

Anscombe's quartet: four data sets with (almost) identical means, variances, correlations and regression lines.

```
anscombe_long <- anscombe %>%
  pivot_longer(everything(), names_to = c(".value", "set"), names_pattern = "(.)(.)")
anscombe_long %>%
  group_by(set) %>%
  summarise(mean_x = mean(x), mean_y = mean(y), sd_y = sd(y), cor = cor(x, y))
```

```
## # A tibble: 4 x 5
##   set   mean_x mean_y  sd_y   cor
##   <chr>  <dbl>  <dbl> <dbl> <dbl>
## 1 1          9   7.50  2.03 0.816
## 2 2          9   7.50  2.03 0.816
## 3 3          9   7.5   2.03 0.816
## 4 4          9   7.50  2.03 0.817
```

```
ggplot(anscombe_long, aes(x, y)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE, formula = y ~ x) +
  facet_wrap(~ set, nrow = 1)
```



Only set 1 is linear; set 2 is a curve, set 3 has one outlier, set 4 is driven by a single point. Visualisation "forces us to notice what we never expected to see" (Tukey, 1977).

There is no fixed procedure: look from several perspectives and levels of detail, start simply, refine. Tufte's principles: above all show the data; avoid distorting what the data have to say; present many numbers in a small space; encourage the eye to compare; reveal the data at several levels of detail.

Main tools: histograms, densities, bar plots, boxplots (distributions); scatterplots (bivariate data); heat maps and maps (specialised).

2.1.1 Example: campaign spending in 2012

US Federal Election Commission spending data, 2012 (slides). Three lessons:

- **A third variable can reverse the story.** Total spending made Obama look best funded, but colouring by type of spending showed that $251 million of the $274 million spent on his campaign was spent *against* him (92%, against 70% for Romney).
- **Faceting adds context.** By office (House, Senate, President): only in the presidential race was more spent attacking Democrats than Republicans.
- **Proportions, not totals.** In House races compare vote share with the *share* of spending (districts differ in turnout). Use a log scale for skewed amounts; overlay or stack distributions to compare them.

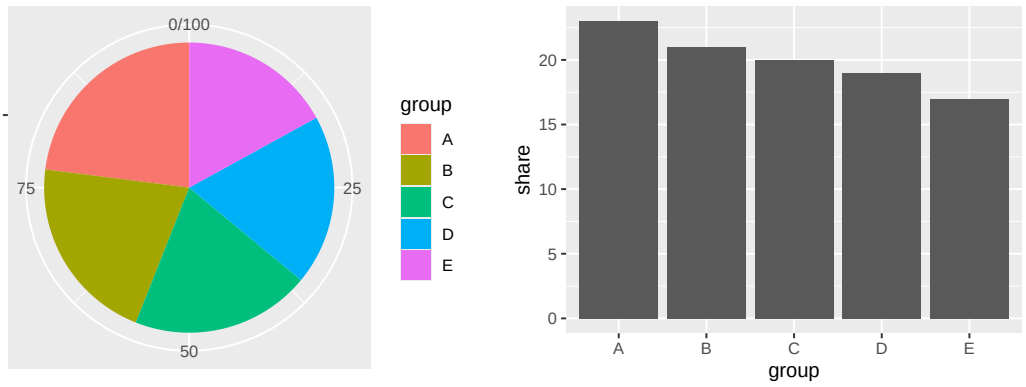
2.2 A taxonomy for data graphics

Motivation. Yau's four elements (Yau, 2013), matching ggplot2's grammar of graphics, break any graphic into parts.

1. **Visual cues**: graphical elements that draw the eye. In decreasing order of how accurately people perceive differences: position, length, angle, direction, shape (categorical), area, volume, shade and colour.
2. **Coordinate system**: Cartesian, polar, or geographic.
3. **Scale**: how distance in the graphic translates into differences in quantity. Numeric scales may be linear, logarithmic or percentage; categorical scales may be unordered or ordinal; time is numeric but has calendar units and can be periodic.
4. **Context**: titles, axis labels, reference points or lines that help the viewer make meaningful comparisons.

Positions and lengths are judged well, angles and areas poorly: **bar charts beat pie charts**; bubble sizes and colour differences convey magnitudes only roughly; red–green fails for colour-blind viewers.

```
shares <- tibble(group = LETTERS[1:5], share = c(23, 21, 20, 19, 17))
ggplot(shares, aes(x = "", y = share, fill = group)) +
  geom_col(width = 1) + coord_polar(theta = "y") + labs(x = NULL, y = NULL)
ggplot(shares, aes(group, share)) + geom_col()
```



Same five shares: equal-looking slices, obvious bar ordering.

Dissecting a graphic. Ask: which variables (and types), which cue for each, which coordinates and scales, what context. E.g. mean SAT score by state: continuous score and categorical state, length cue, Cartesian, linear, axis labels and title.

2.3 A directory of visualisations

Motivation. Most plotting questions fall into a few types, each with a standard plot.

what to show	plots
amounts	bars (vertical or horizontal), dots where the bar would end
one distribution	histogram, density (need a binwidth or bandwidth); empirical CDF, QQ plot (faithful but harder to read)
several distributions	boxplots, violins, overlapping densities, ridgeline plots (many groups)
proportions	side-by-side bars, stacked bars (many conditions), pies (only for simple fractions of a whole); stacked densities when proportions change along a continuous variable
multiple groupings	mosaic plots, treemaps, parallel sets
x–y relationships	scatterplots, bubble charts (third variable as size), 2-d bins for many points, correlograms for many variables, line graphs when x is time
geospatial	choropleth maps, cartograms

2.4 Visualising distributions

Motivation. Histograms and densities depend on a bin width or bandwidth; standard rules give a starting point.

2.4.1 Histograms and bin width

Bin width h: too narrow shows noise, too wide hides structure (slides: Titanic ages, widths 1, 3, 5, 15). Two rules:

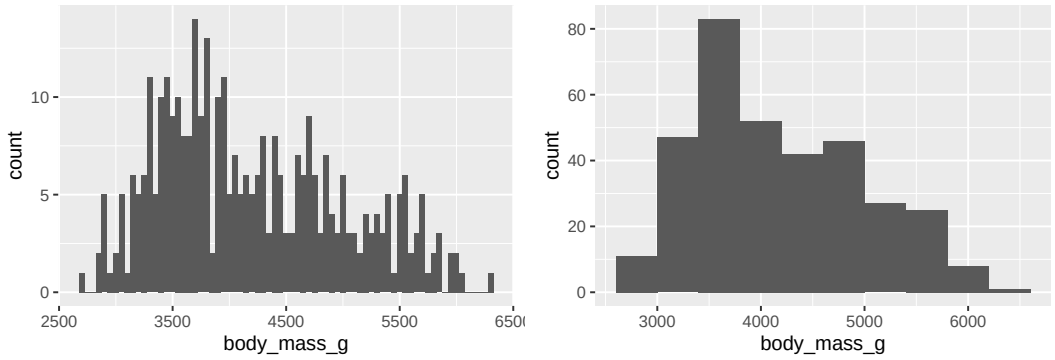
$$\text{Scott: } h = 3.49\,\hat{\sigma}\,n^{-1/3}, \qquad \text{Freedman–Diaconis: } h = 2\,\text{IQR}\,n^{-1/3},$$

the second robust to outliers. Penguin body mass:

```
x <- na.omit(penguins$body_mass_g)
c(n = length(x), scott = 3.49 * sd(x) * length(x)^(-1/3),
  fd = 2 * IQR(x) * length(x)^(-1/3))
```

```
##       n    scott       fd
## 342.0000 400.2209 343.1910
```

```
ggplot(penguins, aes(body_mass_g)) + geom_histogram(binwidth = 50, na.rm = TRUE)
ggplot(penguins, aes(body_mass_g)) + geom_histogram(binwidth = 400, na.rm = TRUE)
```



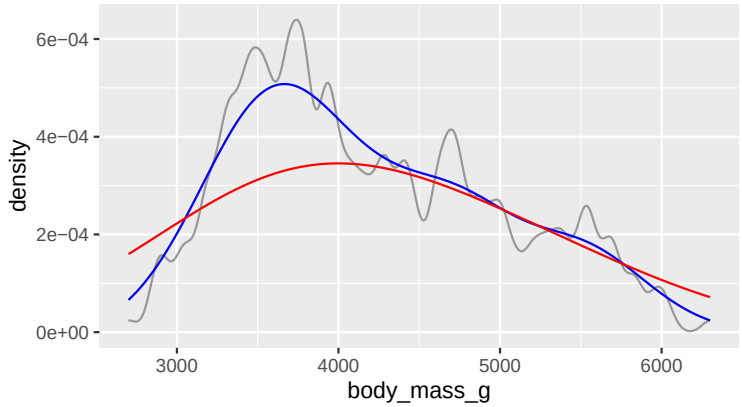
2.4.2 Density estimates

A kernel density estimate sums a bump (**kernel** K) at each observation:

$$\hat{f}(x) = \frac{1}{nh} \sum_{i=1}^{n} K\Big(\frac{x - X_i}{h}\Big).$$

The bandwidth h matters far more than K. Rule of thumb (Härdle, 1991): $h = 1.06\, S\, n^{-1/5}$, where S is the standard deviation; robust versions replace S by the scaled IQR $(X_{(0.75n)} - X_{(0.25n)})/1.34$ or the scaled MAD $1.4826 \cdot \text{median}|X_i - \text{median}(X)|$.

```
ggplot(penguins, aes(body_mass_g)) +
  geom_density(bw = 50, colour = "grey60", na.rm = TRUE) +
  geom_density(bw = 1.06 * sd(x) * length(x)^(-1/5), colour = "blue", na.rm = TRUE) +
  geom_density(bw = 800, colour = "red", na.rm = TRUE)
```



Where the 1.06 comes from [Supplement]

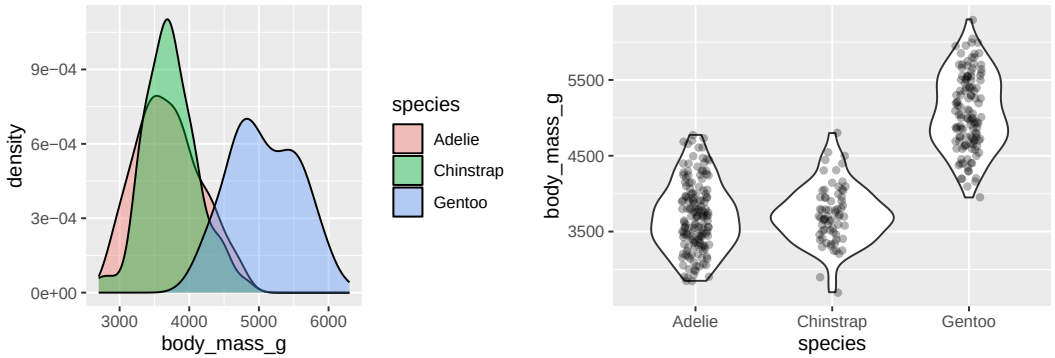
For a normal density with standard deviation σ and a Gaussian kernel, the bandwidth minimising the asymptotic mean integrated squared error is $h^* = (4/3)^{1/5}\sigma n^{-1/5} \approx 1.06\,\sigma n^{-1/5}$. It oversmooths multimodal data (the blue curve blurs the species).

2.4.3 Comparing distributions

Two distributions: overlay (with transparency). Several: jittered points, boxplots, violins, or ridgelines (`ggridges`, many groups).

```
ggplot(penguins, aes(body_mass_g, fill = species)) +
  geom_density(alpha = 0.4, na.rm = TRUE)
```

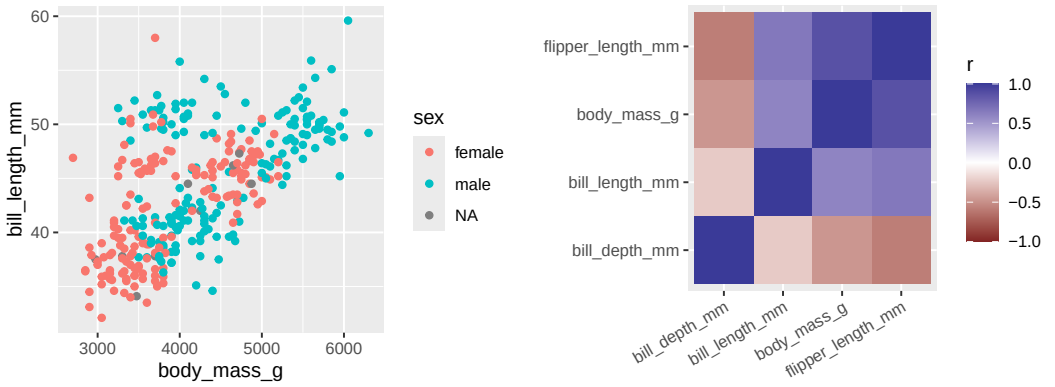
```
ggplot(penguins, aes(species, body_mass_g)) +
  geom_violin(na.rm = TRUE) + geom_jitter(width = 0.1, alpha = 0.3, na.rm = TRUE)
```



2.5 Visualising associations

(Slides: 123 blue jays.) Scatterplot for two variables; colour for a group; size for a third variable (**bubble chart**, hard to judge); scatterplot matrix; **correlogram** (picture of the correlation matrix).

```
ggplot(penguins, aes(body_mass_g, bill_length_mm, colour = sex)) +
  geom_point(na.rm = TRUE)
penguins %>%
  select(bill_length_mm, bill_depth_mm, flipper_length_mm, body_mass_g) %>%
  cor(use = "complete.obs") %>%
  as_tibble(rownames = "x") %>%
  pivot_longer(-x, names_to = "y", values_to = "r") %>%
  ggplot(aes(x, y, fill = r)) + geom_tile() +
  scale_fill_gradient2(limits = c(-1, 1)) +
  theme(axis.text.x = element_text(angle = 30, hjust = 1)) + labs(x = NULL, y = NULL)
```



`GGally::ggpairs()` draws the full scatterplot matrix in one call.

Simpson's paradox in the penguins [Supplement]

Pooled over species, bill depth and bill length are negatively correlated; within each species the correlation is positive. A grouping variable can reverse an association.

```
penguins %>% drop_na() %>%
  summarise(pooled = cor(bill_length_mm, bill_depth_mm))
```

```
## # A tibble: 1 x 1
##   pooled
##    <dbl>
## 1 -0.229
```

```
penguins %>% drop_na() %>% group_by(species) %>%
  summarise(within = cor(bill_length_mm, bill_depth_mm))
```

```
## # A tibble: 3 x 2
##   species   within
##   <fct>      <dbl>
## 1 Adelie     0.386
## 2 Chinstrap  0.654
## 3 Gentoo     0.654
```

2.6 Visualising trends

Motivation. Smoothing separates trend from noise; a transformation can straighten a curved trend.

Defined functional forms. bioRxiv submissions grow as $y = Ae^{mx}$; fit a line to $\log y = \log A + mx$.

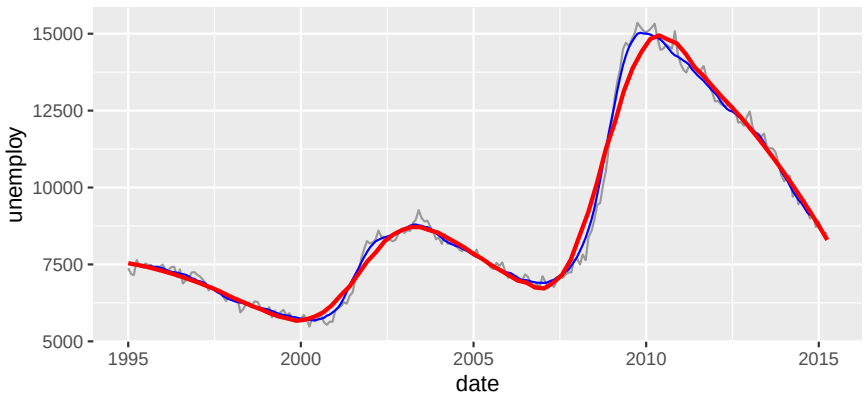
Smoothing. (Slides: Dow Jones 2009.)

- a **moving average**: the mean over a window of neighbouring times;
- **LOESS** (locally estimated scatterplot smoothing): at each x_0, take the k nearest observations $N_k(x_0)$ and fit a line $\ell(x) = \beta_0 + \beta_1(x - x_0)$ by weighted least squares, minimising

$$\sum_{i:X_i \in N_k(x_0)} \{Y_i - \ell(X_i)\}^2 \, W\Big(\frac{|x_0 - X_i|}{\Delta(x_0)}\Big),$$

  where $\Delta(x_0) = \max_{i:X_i \in N_k(x_0)} |X_i - x_0|$ and $W(u) = (1 - u^3)^3$ (tri-cube weight). The **span** k/n controls smoothness; a quadratic can replace the line;
- **splines**: piecewise polynomials joined smoothly at *knots*; faster than LOESS, but sensitive to type and number of knots (Lecture 10).

```
econ <- economics %>%
  filter(date >= as.Date("1995-01-01")) %>%
  mutate(ma12 = stats::filter(unemploy, rep(1/12, 12), sides = 2))
ggplot(econ, aes(date, unemploy)) +
  geom_line(colour = "grey60") +
  geom_line(aes(y = ma12), colour = "blue", na.rm = TRUE) +
  geom_smooth(method = "loess", span = 0.3, se = FALSE, colour = "red", formula = y ~ x)
```



US unemployment (thousands, `ggplot2::economics`): raw series (grey), 12-month moving average (blue) and LOESS with span 0.3 (red).

2.7 Telling a story

Motivation. The Challenger disaster is the classic case of the right data shown the wrong way.

On 27 January 1986 Morton Thiokol engineers urged delaying the Challenger launch, fearing cold would compromise the O-rings; their 13 charts did not persuade NASA, and the O-rings failed 73 seconds after launch. The charts never plotted damage against temperature, and the hearings showed only damaged flights. Tufte's plot of damage against temperature for *all* flights: every launch below about 65°F had damage, and the forecast (about 30°F) was far below every previous launch.

Making a point simply. For the message "American and Delta have the shortest delays", a bar chart beats a cluttered delay-over-time plot; flight counts show both are major NYC carriers:

```
by_carrier <- flights %>%
  group_by(carrier) %>%
  summarise(delay = mean(arr_delay, na.rm = TRUE), n = n()) %>%
  filter(n > 10000)
ggplot(by_carrier, aes(reorder(carrier, delay), delay)) + geom_col() +
  labs(x = "carrier", y = "mean arrival delay (min)")
ggplot(by_carrier, aes(reorder(carrier, delay), n)) + geom_col() +
  labs(x = "carrier", y = "number of flights")
```

